**SDC HACKATHON 2026**

# Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

## 1. Team & Project Overview

| | |
|---|---|
| **Team Name** | Hands Off Our Rights |
| **Project / Solution Name** | Future Skills: find the part of IT that suits you, and the job to aim for |
| **Name (Leader)** | Hong Zhi Yang |
| **Email & Phone (Leader)** | zhiyang19015@gmail.com \| +60 11-4035 9109 |
| **University Affiliation(s)** | Universiti Tunku Abdul Rahman (UTAR) |
| **Live Prototype URL** | https://future-skills.replit.app |
| **Code Repository URL (public / access granted to judges)** | https://github.com/pekoremon/future-skills |

# 2. Architecture & Technology Stack

## 2.1 System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

The browser front end (plain HTML, CSS and JavaScript, no build step; English, Bahasa Melayu and Chinese) opens on the Future Skills Radar, then guides a student through an optional CV upload or skill ratings, a 36-item interest questionnaire and a short skills test on the 3–5 IT areas these point to. PDF CVs are read with PDF.js and Python answers run with Pyodide, both in the browser.

The back end is one Python 3.12 program using only the standard library. It scores the questionnaire and test, matches the student to 25 roles in 5 IT areas, applies three future scenarios, and builds learning paths and SDG contributions from CSV tables in data/ (O*NET, DOSM, MyCOL, arXiv, courses), where every row carries its source and date. SQL answers run in a read-only, in-memory SQLite database.

Only the CV reader and "Why this job?" call an LLM: gpt-oss-120b through an OpenAI-compatible API (Fireworks AI); if the call fails, rule-based text is shown. "Update data" refreshes trend inputs from the OpenDOSM and arXiv APIs. Hosting: Replit Deployments; code: GitHub.

## 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| Python 3.12 standard library (http.server, sqlite3, csv, json, gzip, urllib, unittest) | Back end, data scripts, 172 automated tests; no third-party Python packages | PSF License |
| JavaScript (ES2020), HTML5, CSS3 | Web front end, written by the team without a framework | Open web standards |
| Pyodide 0.26.4 (jsDelivr CDN) | Runs students' Python code answers in the browser | MPL 2.0 |
| PDF.js 3.11.174 (Mozilla, cdnjs CDN) | Reads the text of a PDF CV in the browser | Apache 2.0 |
| SQLite (bundled with Python) | Read-only, in-memory database for SQL test answers | Public domain |
| UN SDG icons | The 17 goal tiles on matches and learning paths | UN SDG icon guidelines (information use) |

# 3. APIs & Cloud Services

## 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| Fireworks AI chat completions API (OpenAI-compatible) | CV reading and the "Why this job?" / "Explain this for me" text with gpt-oss-120b | Paid (team's own account) |
| OpenDOSM / data.gov.my Data Catalogue API (DOSM) | Employment and wage data for the "Update data" trend refresh | Free, no key |
| arXiv API (Cornell University) | AI paper counts per IT area for the AI-momentum signal | Free, no key |
| jsDelivr and cdnjs (public CDNs) | Serve Pyodide and PDF.js to the browser | Free |
| Microsoft Learn Catalog API, YouTube oEmbed (development only) | Collecting course details and checking that every learning link opens (check_links.py) | Free, no key |

## 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Replit Deployments (Replit) | Hosts the live prototype; the LLM key is kept in Replit Secrets | Cloud Run deployment target, Replit default region; SDC sponsor credits |
| Replit Workspace (Replit) | Shared development, preview and test environment; publishing | SDC sponsor credits |
| GitHub (GitHub, Inc.) | Public code repository and commit history | Free plan |
| Fireworks AI serverless inference | Runs the gpt-oss-120b model the app calls | Pay as you go, team's own account |
| Replit web analytics (Replit) | Page-view statistics Replit Deployments adds to the live site | Included with Replit Deployments |

| Service & Provider | Purpose in Project | Plan / Region |
| --- | --- | --- |
| Database, file storage, login | None: data ships as CSV files in the repository; nothing is stored per user on the server | Not applicable |

# 4. AI Tools, Models & Datasets

## 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| gpt-oss-120b (OpenAI open-weight model, 120B) | Fireworks AI (accounts/fireworks/models/gpt-oss-120b) | Pre-trained, used as is (no fine-tuning) | Apache 2.0 with the gpt-oss usage policy; Fireworks AI terms |
| No other model | Matching, test grading, trend scores, scenarios and learning paths are rule-based code written by the team | Not applicable | Not applicable |

## 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| O*NET 31.0 Database (skills, technologies, interests, job zones) | https://www.onetcenter.org/database.html | Public | CC BY 4.0 | N |
| DOSM / OpenDOSM labour, wage, Job Market Insights and Graduates Statistics 2024 | https://open.dosm.gov.my ; https://www.dosm.gov.my | Public | OpenDOSM CC BY 4.0; DOSM releases cited | N |
| MyCOL 2024/2025 critical occupations list (TalentCorp) | https://www.talentcorp.com.my/mycol/ | Public | Published report, cited | N |
| arXiv paper counts per category | https://export.arxiv.org/api/query | Public | arXiv metadata CC0 | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| UN SDG targets (233 role/skill links with reasons) | https://unstats.un.org/sdgs/indicators/indicators-list/ | Public; mapping self-made | UN, cited; mapping Apache 2.0 | N |
| Free course and video links (32 courses, 158 resources: Microsoft Learn, CS50, freeCodeCamp, Cisco, AWS, Google, etc.) | data/courses.csv, data/learning_resources.csv | Public (links and titles only) | Each creator's terms; linked, not copied | N |
| IT skills test (77 questions) and interest questionnaire (36 items) | data/it_assessment.json, data/interest_items.json | Self-made by the team | Apache 2.0 (project licence) | N |
| Demo student profiles | data/sample_profiles.json | Synthetic (fictional people) | Apache 2.0 (project licence) | N |

### 4.3 AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

| Tool & Version | Used For | Extent of Use (minor / moderate / major) |
|---|---|---|
| Claude Code (Anthropic, Claude Opus models) | Writing and testing code, data cleaning and checks, research for sources, drafting documentation | Major |
| ChatGPT (OpenAI) | Discussing ideas and answering questions in chat; not used to write code | Minor |
| Replit Agent (Replit) | Deployment settings and publishing on Replit | Minor |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

The team set the direction, the IT areas, the questionnaire items and the score weights; Claude Code wrote code to that plan. Every change was run before merging: 172 automated tests, a data validator that checks each row has a source and date, a link checker, and a walk-through in the browser on desktop and phone. Data the AI gathered was checked against the original sources. Team members reviewed the code and can explain each part; no number in the app is invented.

## 5. Sponsor-Provided Resources

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

| Sponsor | Resource (API / credits / platform / data) | How It Was Used in the Project |
|---|---|---|
| Replit | Cloud credits and platform (Workspace and Deployments) | Development workspace, preview and hosting of the live prototype |

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

Replit hosts the live prototype, but the app does not depend on it: it is one Python program with no packages to install, so it runs anywhere with Python 3.12 (python src/main.py). No sponsor GenAI API is used by the deployed app; the Fireworks AI key is the team's own. The code accepts any OpenAI-compatible endpoint, so the KrackedDevs sponsor endpoint could replace it by changing one secret.

## 6. Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

No accounts or logins, and no analytics of our own; Replit Deployments adds its standard page-view script. Answers, skill ratings and learning progress stay in the student's browser (localStorage) and are sent to our server only to compute results; the server keeps nothing.

If a student uploads a CV, its text goes to our server and to Fireworks AI to find evidence per IT area. Our server never writes it to disk, logs or caches it, and the page asks students to leave out their IC number, address and phone. "Why this job?" sends only computed match facts. The API key is in Replit Secrets, not in the code. Demo profiles are fictional. The app collects no age, gender or background.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

Hallucination: the model only words text. A reply with any number not in our computed facts is rejected and a rule-based text shown; CV quotes not found in the CV are dropped.

Single provider: if Fireworks AI fails, every feature still works with fallback text, and any OpenAI-compatible API can replace it.

Data: role skills come from O*NET (U.S.). Malaysian per-role demand data is thin (DOSM publishes job ads for one IT occupation; MyCOL is one edition), so trend scores are coarse and labelled as estimates; each number shows its source and date, a missing number is never estimated, and a failed source keeps its last good data.

Bias: weights are team choices shown on screen; scores never hide a role.

## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Team leader: Hong Zhi Yang  Signature: _____

Date: 1 November 2026